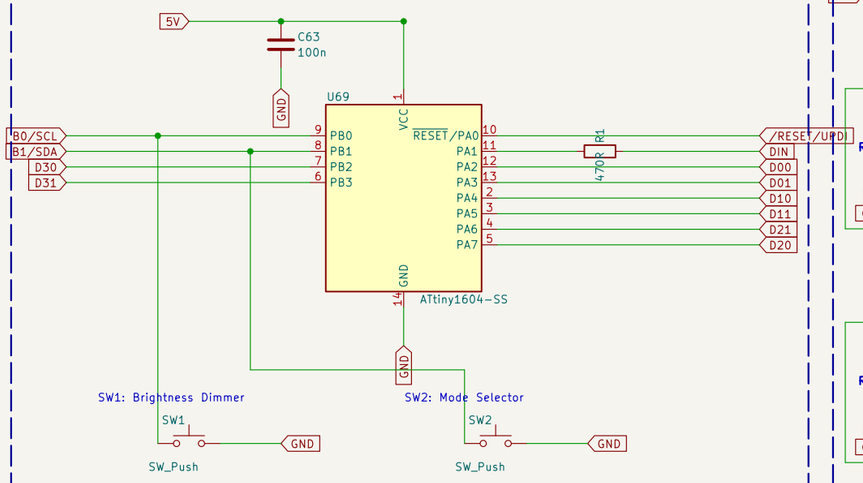
1. MICROCONTROLLER & CONTROLS (ATtiny1604)

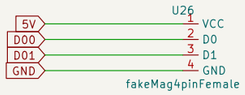
20MHz AVR Core | Dual User Buttons | In-Line R1 Protection | C63 Bypass Cap



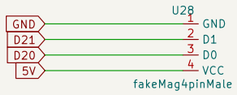
2. 4-WAY MAGNETIC SNAP BUS CONNECTORS

Self-Aligning Edge Pogo Connectors with Instant Hardware Detection

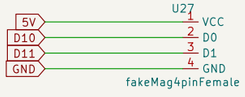
NORTH EDGE (Female)



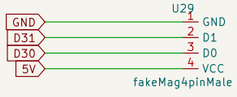
EAST EDGE (Male)



SOUTH EDGE (Female)



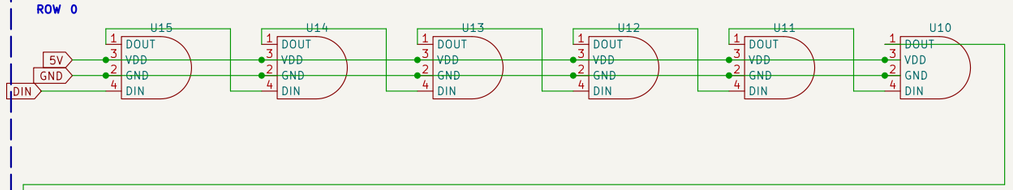
WEST EDGE (Male)



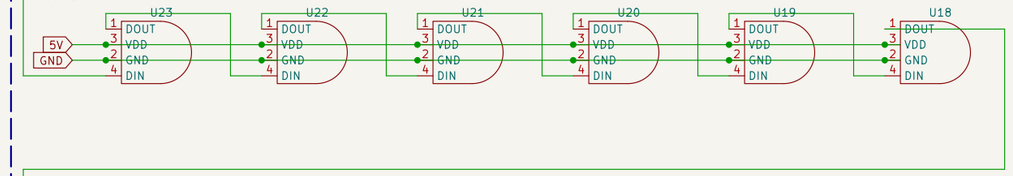
3. 6x6 WS2812D ADDRESSABLE RGB LED MATRIX

Continuous Visible Serpentine Chain (Row 0 -> Row 5) | 36x WS2812D Pixels

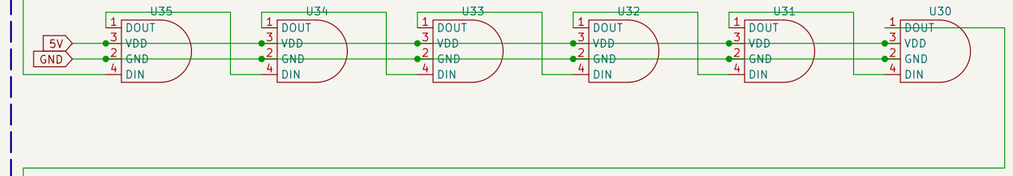
ROW 0



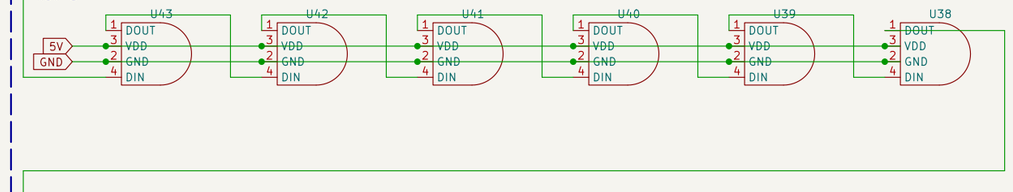
ROW 1



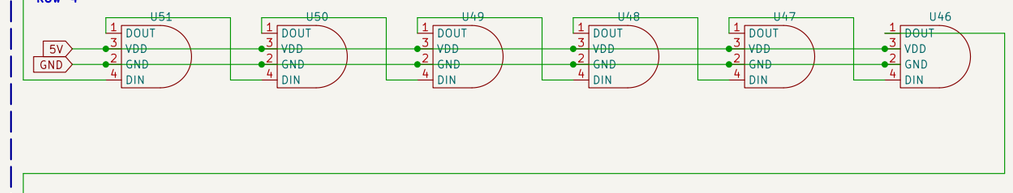
ROW 2



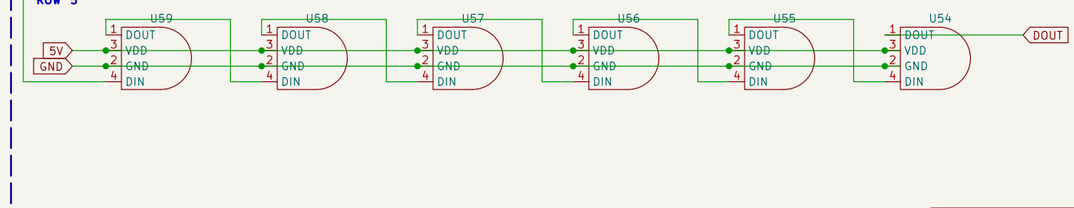
ROW 3



ROW 4



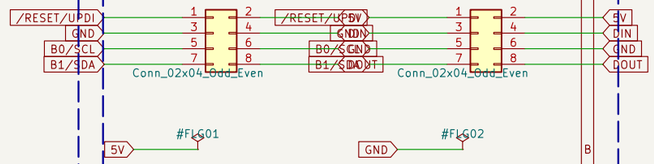
ROW 5



4. PROGRAMMING & EXPANSION HEADERS

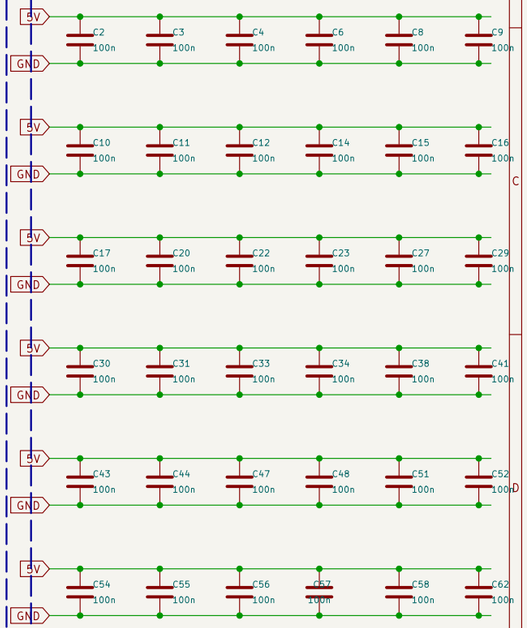
Dual 2x4 SMD Headers for 1-Wire UPDI Flashing & Bus Mirroring

J1 (UPDI & EXPANSION) **J2 (MIRROR BUS)**



5. DECOUPLING CAPACITORS (36x 100nF)

Dedicated 100nF 0603 Ceramic Bypass per WS2812D LED (Pin 1=5V, Pin 2=GND)



6. SYSTEM ARCHITECTURE & TOPOLOGY

* Complementary Pogo Gender: North/South Female, East/West Male
* Neighbor Detection: D00..D31 form complementary pairs across tiles
* Firmware: megaTinyCore in Arduino IDE (ATtiny1604, 20MHz, 800kHz)

Clean Modular Schematic Layout
Serpentine Daisy Chain with Complementary Neighbor Sensing
6x6 WS2812D Addressable LEDs + ATtiny1604-SS + 4-Way Magnetic Snap Bus
Antigravity Hardware

Sheet: /
File: modularmatrix.kicad_sch

Title: Modular LED Matrix 6x6 (Bitluni Architecture)

Size: A3	Date: 2026-11-01	**Rev: v2.0-aligned**
KiCad E.D.A. 10.0.3		Id: 1/1